PyQED: An Open Source Quantum Electrodynamic Correction Post-Processing Tool for Quantum Chemical Calculations

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We present PyQED, an open-source Python post-processing tool that computes Uehling vacuum polarization corrections from converged ORCA quantum chemical calculations. PyQED interfaces through `orca_2json` to extract basis set definitions and the SCF density matrix, then evaluates the screened finite-nucleus Uehling potential on a Becke-Lebedev integration grid, and returns the QED energy correction perturbatively as $\Delta E = \mathrm{Tr}(\mathbf{P}\cdot\mathbf{V}^{\mathrm{Ueh}})$. The Uehling potential is the leading vacuum polarization contribution to QED corrections; the electron self-energy, which is larger in magnitude but opposite in sign, is not included in the current implementation but is planned. We validate on thorium and uranium atoms and apply the tool to actinide fluorides (ThF_4, ThF_6^{2-}, ThF_7^{3-}, UF_6). Total Uehling corrections are large (-249 eV for Th and -287 eV for U) but are mainly by core s-orbitals and essentially constant across chemical environments; the thorium contribution varies by less than 0.005 eV from ThF_4 to ThF_7^{3-}. Each fluorine contributes only 10.6 meV. Therefore, QED corrections nearly vanish in energy differences; the net contribution to the ThF_4 atomization energy is 0.11 kcal/mol; including the omitted electron self-energy (estimated at 4-5× the VP magnitude with opposite sign), the total QED contribution to reaction energetics is expected to remain below 1 kcal/mol, 50-200× smaller than typical DFT functional errors. These results quantitatively confirm that vacuum polarization corrections are negligible for molecular actinide thermochemistry, while providing an accessible tool for verification. PyQED is written in Python 3.8+ with NumPy and SciPy dependencies and is available under the MIT license.

I. INTRODUCTION

Modern quantum chemistry packages (ORCA, Gaussian, Psi4, Q-Chem) handle relativistic effects well. They use scalar relativistic Hamiltonians (ZORA, Douglas-Kroll-Hess), spin-orbit coupling, and various correlation methods that produce reliable results for heavy elements. What they don't include is quantum electrodynamics. Specifically, vacuum polarization and electron self-energy corrections are absent from standard methods. Of course, specialized codes exist like GRASP[1], DIRAC[2] and BERTHA[3]. The latter two use four-component methods. These are excellent tools for benchmark calculations requiring sub-eV accuracy. But if you simply want to check whether QED matters for your actinide complex, learning an entirely new computational framework is usually overkill. This gap between "ignore QED completely" and "master four-component relativistic quantum mechanics" is what PyQED addresses.

Vacuum polarization comes from virtual electron-positron pairs appearing in and out of existence, and in the presence of a heavy nucleus, they polarize like a dielectric medium. The virtual electron gets pulled toward the nucleus, the virtual positron gets pushed away, and the net effect is a small screening of the nuclear charge. The leading contribution is the Uehling potential [4]. For element 111 ($Z = 111$), the total 1s self-energy including screening and finite nuclear size is approximately 779 eV, while the valence 7s contribution is 0.08 eV [5]. Vacuum polarization is typically an order of magnitude smaller than the self-energy but follows the same $(Z\alpha)^4$ scaling with nuclear charge. QED corrections scale as $(Z\alpha)^4$ [6], where $\alpha \approx 1/137$ is the fine structure constant, and are therefore concentrated on the heavy atom in a molecule, so light ligands such as fluorine contribute only millielectronvolts each. An important note. This initial version of PyQED evaluates only the vacuum polarization (Uehling) contribution, not the electron self-energy. For heavy atoms, the self-energy is approximately 4-5 times larger in magnitude than the VP and opposite in sign,[5, 7] meaning the net QED correction to total atomic energies is positive and dominated by the self-energy. PyQED therefore cannot determine the sign or magnitude of the total QED correction to absolute energies. However, both VP and SE are core-electron properties that transfer unchanged between chemical environments, so both contributions vanish in energy differences to the same degree. The VP component alone is sufficient to establish whether QED corrections matter for a given reaction energy, which is the intended use case.

QED corrections, however, are tiny compared to other uncertainties. For a typical actinide calculation, basis set incompleteness contributes 10-100 kcal/mol of error, choice of density functional adds 5-20 kcal/mol, and QED contributes 0.001-0.1 kcal/mol. As Neese put it in a private communication, "How important is it to add a 1 cm^{-1} correction if you have hundreds or thousands of wavenumbers of error in other parts of the calculation?" For most chemistry, the answer is that it's not important; QED won't change any conclusions about bonding, structure, or reactivity. This doesn't make QED assessment worthless, however. Knowing something is negligible is more satisfying than assuming it, but be-

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ing honest about scope, this is a tool for evaluation, not a missing piece of computational chemistry. PyQED is a Python post-processor that computes Uehling potential corrections from converged ORCA calculations. It interfaces through `orca_2json` (following Neese's recommendation and the extent of ORCA's 2json development [8]), reads the density matrix and basis set information, evaluates the screened Uehling potential on a numerical grid, and outputs an energy correction. We validate on thorium and uranium atoms, compare to literature benchmarks, and apply it to actinide fluorides (ThF_4, UF_6) to confirm that QED is indeed negligible for chemistry. The code is open-source (MIT license), written in Python 3.8+ with NumPy and SciPy, and is available on GitHub `https://github.com/gbogolyubov/PyQED`. Section 2 describes the theoretical framework and numerical implementation. Section 3 covers computational details. Section 4 presents validation results and molecular applications. Section 5 discusses when QED assessment adds value and compares PyQED to existing tools. Section 6 concludes.

II. METHODS

A. Uehling Potential

The Uehling potential describes vacuum polarization to leading order in QED. For a point nucleus with charge Z, the potential at radius r is

$$V_{\text{Ueh}}(r) = -\frac{2\alpha Z}{3\pi r}\int_1^\infty e^{-2rt/\alpha}\left(1+\frac{1}{2t^2}\right)\sqrt{t^2-1}\,dt \tag{1}$$

where $\alpha \approx 1/137.036$ is the fine structure constant and atomic units are used throughout ($\hbar = m_e = e = 4\pi\epsilon_0 = 1$). The integration variable t represents the ratio of virtual photon momentum to electron mass. The lower limit $t = 1$ corresponds to the pair creation threshold, and the exponential gives us a cutoff at the Compton wavelength $\lambda_C = \alpha \approx 0.0073$ a.u. This explains why core electrons experience much larger QED corrections than valence electrons.

B. Finite Nuclear Size

For heavy nuclei, the point-charge approximation breaks down. The RMS nuclear radius for actinides is roughly 7 fm $\approx 1.3 \times 10^{-4}$ a.u., comparable to the Bohr radius of 1s electrons ($a_0/Z \approx 0.01$ a.u. for thorium). We model the nucleus as a Fermi distribution,

$$\rho_{\text{nuc}}(r) = \frac{\rho_0}{1+\exp[(r-c)/a]} \tag{2}$$

where c is the half-density radius, $a \approx 0.523$ fm is the skin thickness (the diffuseness parameter, related to the 10%-90% skin thickness t by $t = 4a\ln 3$), and ρ_0 is fixed by normalization. Parameters for each element are taken from Angeli and Marinova [9]. The finite-nucleus Uehling potential is found by convolution,

$$V_{\text{Ueh}}^{\text{finite}}(r) = \int_0^\infty \rho_{\text{nuc}}(r')V_{\text{Ueh}}^{\text{point}}(|r-r'|)\,4\pi r'^2\,dr' \tag{3}$$

where spherical symmetry of both the nuclear charge distribution and the Uehling potential has been assumed, reducing the three dimensional convolution to a one-dimensional radial integral. This reduces the 1s correction by roughly 10-20% for actinides, with negligible effect on higher orbitals.

C. Many-Electron Screening

In many-electron systems, inner electrons screen the nuclear field seen by outer electrons. This screening applies to vacuum polarization as well. We use the Welton model,[10] which multiplies the Uehling potential by a screening function,

$$V_{\text{Ueh}}^{\text{scr}}(r) = V_{\text{Ueh}}^{\text{finite}}(r)\cdot\left(\frac{Z-N_{\text{inner}}(r)}{Z}\right) \tag{4}$$

where $N_{\text{inner}}(r) = \int_0^r 4\pi r'^2\rho_e(r')dr'$ is the number of electrons within radius r. The electron density $\rho_e(r)$ is taken directly from the ORCA calculation. At the nucleus, $N_{\text{inner}} \to 0$ so the screening factor approaches 1. Far from the nucleus, $N_{\text{inner}} \to N$ (total electrons) and the screening becomes maximal. This is approximate; a more rigorous treatment would require self-consistent inclusion in the SCF, but it is adequate given that QED corrections are small.

D. Energy Correction

The QED contribution to the total energy is evaluated perturbatively. For a converged wavefunction with density matrix $\mathbf{P}$ in the atomic orbital basis, the energy correction is

$$\Delta E_{\text{QED}} = \sum_{\mu\nu} P_{\mu\nu}V_{\mu\nu}^{\text{Ueh}} \tag{5}$$

where $V_{\mu\nu}^{\text{Ueh}} = \int \chi_\mu^*(r)V_{\text{Ueh}}^{\text{scr}}(r)\chi_\nu(r)d^3r$ are matrix elements of the screened Uehling potential. This treatment neglects orbital relaxation (the orbitals don't adjust to the presence of V_{Ueh}). The approximation is justified because QED corrections are typically $\Delta E/E \sim 10^{-5}$ of the total electronic energy, making second-order effects negligible.

E. Numerical Implementation

Evaluating Eq. 5 requires three levels of numerical integration. First, the Uehling integral (Eq. 1) has an integrable singularity at $t = 1$ and an exponential tail. We

use the substitution $u = \sqrt{t-1}$ to remove the singularity, giving smooth integrands which can be evaluated using Gauss-Kronrod adaptive integration. Integration extends to $u_{\max} = \sqrt{19}$ (corresponding to $t_{\max} = 20$), beyond which the exponential is negligible. Second, the finite nucleus convolution (Eq. 3) is handled by Gauss-Legendre integration on $[0, R_{\text{cutoff}}]$ with $R_{\text{cutoff}} = 5c$ where the nuclear density has fallen below 10^{-6}. Third, matrix elements $V_{\mu\nu}^{\text{Ueh}}$ are found on a 3D grid using Becke atomic partitioning [11] (used for molecular systems; for isolated atoms, no partitioning is required and the grid is centered on the single nucleus) with radial grids (usually 150-200 points) and Lebedev angular grids (order 23-29, corresponding to 194-302 points per shell). Convergence is tested by increasing grid density until energy changes are below 0.1 meV.

F. ORCA Interface

PyQED reads all necessary data from ORCA via the `orca_2json` utility. Following Neese's recommendation, we avoid parsing binary `.gbw` files directly, as this makes the files fragile when changing format. The `orca_2json` tool converts ORCA output into standardized JSON format, from which we extract: (1) nuclear coordinates and atomic numbers, (2) basis set definitions (primitive Gaussians, contraction coefficients), and (3) the SCF density matrix. A typical workflow would look something like this: run ORCA to convergence, call `orca_2json output.gbw output.json`, then use PyQED on the JSON file. Post-processing typically takes <5 minutes depending on system size, negligible compared to the ORCA SCF.

III. COMPUTATIONAL DETAILS

A. ORCA Calculations

All electronic structure calculations used ORCA 6.1.0 [8] with the PBE0 [12] hybrid density functional and ZORA scalar relativistic Hamiltonian. TightSCF convergence criteria were applied. Basis sets were specified explicitly using `NewGTO` blocks because of keyword incompatibilities in this ORCA installation. For thorium, we used the SARC-ZORA all-electron basis (21s, 13p, 10d, 7f contracted shells, 159 functions total). For fluorine, the SARC-ZORA basis has 7s, 4p, 3d, 2f, 1g contracted shells (57 functions). Auxiliary basis sets for Coulomb fitting were `def2/J` for fluorine and `SARC/J` for thorium. Systems studied include bare Th^{4+} and F^- ions, plus molecular species ThF_4 and UF_6. Converged density matrices were exported using `orca_2json` with the configuration {`"Densities": ["scfp"], "Basisset": true`}. Input coordinates were specified in Ångström and converted internally to Bohr for PyQED processing.

B. PyQED Settings

The radial integration grid used 175 points per atom with Mura-Knowles logarithmic mapping, $r = -R\ln(1-x^3)$, where the scaling parameter $R = 3.5$ for actinides. Angular grids used Lebedev integration: order 25 (230 points) for the main integration region, order 11 (50 points) for $r < 0.01$ Bohr and $r > 15$ Bohr, and order 17 (110 points) for the intermediate range $0.01 < r < 0.1$ Bohr. The Uehling integral used the substitution $u = \sqrt{t-1}$ with SciPy's adaptive Gauss-Kronrod integration, integrating to $u_{\max} = \sqrt{19}$ (equivalent to $t_{\max} = 20$) with relative tolerance 10^{-12} and absolute tolerance 10^{-15}. Finite nuclear size was treated using Fermi charge distributions with skin thickness $a = 0.523$ fm and half-density radii from Angeli and Marinova,[9] convoluted using 80-point Gauss-Legendre integration. Many-electron screening followed the Welton model, $V_{\text{scr}}(r) = V_{\text{Ueh}}(r) \times [Z - N_{\text{inner}}(r)]/Z$, with electron density taken from the ORCA SCF density matrix. For molecular systems, Becke atomic partitioning [11] was used with Bragg-Slater radii [13] for size adjustment. The final energy correction was evaluated as $\Delta E = \text{Tr}(\mathbf{P} \cdot \mathbf{V}^{\text{Ueh}})$ in first-order perturbation theory without orbital relaxation. Grid convergence was verified by varying radial and angular grid parameters independently; results are summarized in Table III.

IV. RESULTS

A. Atomic Validation - Thorium

TABLE I. QED Uehling corrections for thorium atom orbitals. PyQED values are orbital energy shifts $\langle i|V_{\text{Ueh}}|i\rangle$ using point-nucleus ZORA wavefunctions.

Orbital	PyQED (eV)
1s	−109.36
2s	−12.22
3s	−2.71
4s	−0.71
5s	−0.17
6s	−0.036
2p	−0.150
3p	−0.039
4p	−0.010
3d	< 0.001
4d	< 0.001
4f	< 0.001
Total	−249.30

The ZORA framework for evaluating Uehling corrections has been independently validated by Janke *et al.*,[14] who showed agreement between two-component

ZORA and four-component Dirac-Hartree-Fock VP corrections to within 0.1-2.6% across the periodic table up to $Z = 90$. To benchmark PyQED directly, we computed per-orbital Uehling corrections for mercury ($Z = 80$) using ZORA wavefunctions. Our 6s vacuum polarization expectation value of -5.20×10^{-3} eV at the Hartree-Fock level agrees to within 0.3% with the VP contribution to the Hg ionization energy of -5.214×10^{-3} eV reported by Janke *et al.*[14] using ZORA-HF. With PBE0 wavefunctions, the value shifts to -5.84×10^{-3} eV, a 12% increase attributable to the larger near-nucleus electron density produced by hybrid DFT. This functional dependence does not affect the paper's conclusions, as the same functional is used consistently for all systems and the QED correction vanishes in energy differences. We note that Janke *et al.* did not systematically explore functional dependence of VP corrections, so the full magnitude of this uncertainty across different density functionals remains unquantified. For this paper's purpose of assessing whether QED is negligible, the 12% functional sensitivity is immaterial compared to the 50-200× gap between QED and other error sources. This is consistent with the observation by Janke *et al.* that basis set and integration grid choices had a larger impact on VP corrections than the level of theory, though functional dependence was not systematically explored. Their ZORA VP corrections agree with four-component Dirac-Hartree-Fock results to within 0.1-2.6% across the periodic table, confirming that the ZORA approximation introduces negligible error in the Uehling potential evaluation for heavy elements. Direct comparison against four-component codes for neutral Th or U was not performed in this work. GRASP, the standard atomic QED code, fails to converge for configurations involving 5d and 6d shells in neutral actinides, making benchmarking impractical for these systems. DIRAC can in principle compute molecular VP corrections but requires substantial expertise and infrastructure beyond the scope of this project. We note, however, that the methodology itself (Uehling VP evaluated perturbatively over ZORA wavefunctions) has been validated to 0.1-2.6% against four-component results across the periodic table by Janke *et al.*,[14] and our Hg 6s benchmark confirms that PyQED reproduces their values to 0.3% at the same level of theory. A systematic comparison against four-component VP corrections for neutral actinides would be desirable and is a priority for future work.

The 1s orbital shows a QED correction of -109 eV, consistent with $(Z\alpha)^4$ scaling expectations for $Z = 90$. Corrections decrease quickly with principal quantum number: 2s contributes -12.2 eV, 3s contributes -2.7 eV, and the pattern continues with each successive s-shell contributing roughly an order of magnitude less. S-orbitals dominate entirely due to their finite density at the nucleus; p-orbitals contribute at most -0.15 eV (2p), while d- and f-orbitals are basically zero. The total QED correction summed over all doubly-occupied orbitals is -249.30 eV (-9.162 Hartree). These values use point-nucleus ZORA wavefunctions; finite nuclear size and many-electron screening reduce the total by approximately 1%.

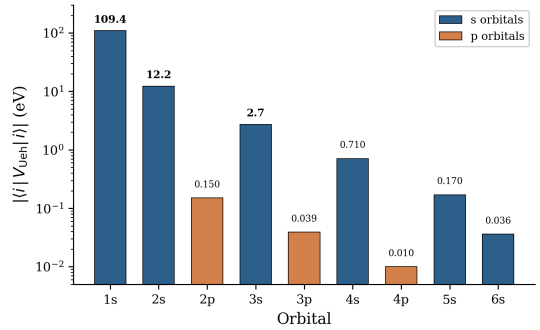

FIG. 1. QED orbital energy shifts for thorium atom on a logarithmic scale. S-orbitals (blue) dominate; p-orbitals (orange) contribute minimally. D- and f-orbitals are below 10^{-3} eV and not shown.

B. Atomic Validation - Uranium

The uranium atom ($Z = 92$) was computed with the same methods. The total QED correction is -286.89 eV (-10.543 Hartree), compared to -249.30 eV for thorium. The ratio of 1.151 is consistent with the expected $(Z_{\rm U}/Z_{\rm Th})^4 = (92/90)^4 = 1.091$ scaling, with the slightly larger ratio coming from contributions by the additional two electrons in uranium occupying s-type orbitals.

C. Molecular Systems

TABLE II. QED Uehling corrections for actinide fluoride molecules. Per-atom decomposition shows QED is dominated by the central actinide. All values include screening and finite nuclear size corrections. Per-F values are per fluorine ligand and are identical across all thorium species to within grid uncertainty.

System	Total (eV)	Metal (eV)	Metal (%)	Per-F (meV)	$N_{\rm F}$
Th atom	-249.299	-249.299	100.0	-	0
ThF_4	-249.337	-249.294	100.0	-10.64	4
ThF_6^{2-}	-249.359	-249.295	100.0	-10.64	6
ThF_7^{3-}	-249.369	-249.295	100.0	-10.64	7
U atom	-286.888	-286.888	100.0	-	0
UF_6	-286.953	-286.889	100.0	-10.65	6

The molecular decomposition reveals two key findings. First, the actinide atom contributes $> 99.99\%$ of the total QED correction in every system; each fluorine contributes only 10.6 meV (0.24 kcal/mol). Second, the actinide contribution is remarkably constant across chemical environments: thorium contributes 249.294 ± 0.005 eV

whether in ThF_4, ThF_6^{2-}, or ThF_7^{3-}. We note that the grid convergence spread of ∼0.03 eV (Table III) was determined for Th^{4+}; the ±0.005 eV molecular variation is smaller than this absolute uncertainty but is systematic (monotonically increasing with N_F) rather than random, consistent with a small but real perturbation of the core density by the ligand field. The insensitivity to coordination number and charge state reflects the fact that QED corrections arise from electron density within ∼ 0.007 Bohr of the nucleus, where the wavefunction is dominated by core orbitals unaffected by chemical bonding. The practical consequence for reaction energetics is that QED corrections nearly vanish in energy differences. For the atomization reaction $ThF_4 \rightarrow$ Th +4F, the net QED contribution is: For the atomization reaction $ThF_4 \rightarrow$ Th +4F, the net QED contribution is:

$$\Delta E_{\mathrm{QED}} = E_{\mathrm{QED}}(\mathrm{ThF_4}) - E_{\mathrm{QED}}(\mathrm{Th}) - 4\,E_{\mathrm{QED}}(\mathrm{F}) \quad (6)$$

which evaluates to +0.005 eV (0.11 kcal/mol). This is 50-200× smaller than typical DFT functional errors (5-20 kcal/mol) and is comparable to numerical noise in the integration grid.

D. Grid Convergence

TABLE III. Grid convergence of the total QED correction for Th^{4+} (point nucleus, unscreened). Angular convergence is rapid due to the spherical symmetry.

Radial convergence (angular order 25)		
Radial points	QED (Hartree)	Δ from 250-pt (meV)
100	−9.24509	−8.4
150	−9.24736	−2.3
175	−9.24918	+2.6
200	−9.24790	+0.8
250	−9.24819	-
Angular convergence (175 radial points)		
Lebedev order	QED (Hartree)	Δ from order 29 (meV)
11 (50 pts)	−9.24918	< 0.01
17 (110 pts)	−9.24918	< 0.01
23 (194 pts)	−9.24918	< 0.01
25 (230 pts)	−9.24918	< 0.01
29 (302 pts)	−9.24918	-

Grid convergence was verified by varying radial and angular grid parameters independently. For angular covergence, all Lebedev orders from 11 to 29 give identical results to all reported digits, consistent with the spherical symmetry of the dominant s-orbital contributions. Radial convergence shows sub-eV variation across grid sizes, with a spread of approximately 0.03 eV (∼0.01% of the total) between 100 and 250 radial points. The non-monotonic behavior reflects the sensitivity of the Mura-Knowles logarithmic mapping to the placement of grid points near the nuclear cusp. The 175-point grid gives us a result within 0.03 eV of the 250-point value, and the spread across all grid sizes (100-250 points) is 0.1% of the total correction. We use 175 points as a practical compromise between accuracy and computational cost.

E. Basis Set Convergence

TABLE IV. Basis set convergence of total QED correction for Th atom.

Basis Set	QED (eV)	Δ from QZ (eV)
SARC-ZORA-DZ	−249.2991	< 0.001
SARC-ZORA-QZ	−249.2991	-

The QED correction is identical at DZ and QZ basis set levels to within 0.001 eV, a complete basis set convergence. This is expected because the Uehling potential is confined to $r < \lambda_C \approx 0.007$ Bohr from the nucleus, a region where even minimal basis sets accurately represent the core electron density. The tight s-type primitives with exponents exceeding 10^7 that describe the 1s orbital are present in both DZ and QZ basis sets with identical contraction coefficients. Practically, this means that QED corrections can be computed at minimal basis set cost without sacrificing accuracy. We emphasize that this rapid convergence is specific to the Uehling correction, which only looks at the innermost electron density; the electronic energy itself requires much larger basis sets for convergence.

F. QED in Error Budget

TABLE V. Error hierarchy for actinide fluoride calculations. QED (per atom) is the total absolute Uehling correction for the Th atom, not a computational error; it is included to show the scale of cancellation in energy differences.

Source	kcal/mol	vs. QED
Basis set	10–100	100–1000×
DFT functional	5–20	50–200×
ZORA vs. DKH	1–5	10–50×
Spin-orbit	1–10	10–100×
QED (net, rxn)	∼0.1	1×
QED (per F)	0.24	—
QED (per atom)	5749	—

Interestingly, QED corrections are enormous in absolute terms (∼5750 kcal/mol for the thorium atom) but are negligible for chemistry (∼0.1 kcal/mol for reaction energies). This cancellation happens because QED is a property of the atomic core that transfers unchanged

between chemical environments. Every source of error in standard computational chemistry (basis set incompleteness, density functional approximation, scalar vs. two-component relativistic treatment, and spin-orbit coupling) exceeds the net QED contribution by at least an order of magnitude. As Neese noted in a private communication, adding a 1 cm^{-1} correction when other errors are hundreds or thousands of wavenumbers is not chemically meaningful. Our results quantify this though; the QED correction to the ThF_4 atomization energy is 0.11 kcal/mol, while the DFT functional error alone is 5-20 kcal/mol. For practical actinide thermochemistry, QED can be safely neglected.

V. DISCUSSION

A. When QED Matters vs. Doesn't

These results establish a framework to decide when QED corrections require evaluation in actinide calculations. The total Uehling correction scales as approximately $(Z\alpha)^4$ and reaches hundreds of electronvolts for heavy elements: -249 eV for thorium and -287 eV for uranium. These are large numbers by any measure. However, the correction is almost entirely a property of the atomic core. Across ThF_4, ThF_6^{2-}, and ThF_7^{3-}, the thorium contribution varies by only ± 0.005 eV despite dramatic changes in coordination number and formal charge. Each fluorine ligand contributes a fixed -10.6 meV regardless of the molecule. The consequence is that the dominant core QED contribution transfers unchanged between chemical environments, leaving only a small residual in energy differences. Formation energies, bond dissociation energies, reaction barriers, and isomerization energies all involve differences between systems containing the same heavy atom, so the dominant atomic QED contribution drops out. The residual is on the order of $N_{\mathrm{F}} \times 0.01$ eV, which for ThF_4 amounts to 0.11 kcal/mol.

QED assessment is therefore unnecessary when: (1) the quantity of interest is an energy difference between species containing the same actinide, (2) the target accuracy is above 1 kcal/mol, or (3) other error sources (basis set, functional, spin-orbit treatment) exceed 1 kcal/mol. In practice, this covers essentially all molecular actinide calculations performed today. However, QED assessment may become relevant when: (1) sub-kcal/mol accuracy is required for benchmark thermochemistry, (2) comparing total energies of systems with different heavy elements, or (3) computing core-level spectroscopic properties such as X-ray binding energies where the absolute orbital energy matters.

B. Comparison to Existing Tools

Several codes compute QED corrections with high accuracy. GRASP[1] is a multiconfigurational Dirac-Hartree-Fock atomic structure code that includes radiative QED corrections self-consistently; it is essentially the gold standard for atomic calculations but is limited to atoms and requires expertise in relativistic atomic physics. DIRAC[2] provides four-component relativistic molecular calculations and can evaluate QED corrections at the Dirac-Coulomb level, but the computational cost and methodological complexity are substantial. BERTHA[3] similarly operates in the four-component framework. PyQED is not intended to compete with these codes on accuracy; it uses ZORA wavefunctions rather than four-component Dirac spinors, treats QED perturbatively rather than self-consistently, and currently includes only the Uehling contribution. However, it is accessible; a researcher running standard ORCA calculations on actinide systems can evaluate QED corrections as a post-processing step without learning a new computational framework, changing their workflow, or performing additional SCF calculations. The entire computation takes seconds to minutes and requires only the converged `.gbw` file. For the purpose of confirming that QED is negligible for a particular application, this level of theory is sufficient.

C. Limitations

PyQED computes only the Uehling (vacuum polarization) contribution to QED corrections. The larger correction by magnitude comes from electron self-energy corrections, which are more difficult to evaluate and are deferred to future work. It is perturbative - the Uehling potential is evaluated as a first-order expectation value over converged ORCA orbitals, and neglects orbital relaxation in response to the QED potential. This approximation is justified by the small magnitude of $\Delta E_{\mathrm{QED}}/E_{\mathrm{total}} \sim 10^{-5}$. The use of ZORA wavefunctions rather than four-component Dirac spinors introduces some error in the near-nucleus electron density, particularly for the 1s orbital. Finally, the many-electron screening is treated using the Welton model rather than self-consistent inclusion in the SCF procedure. The Welton model has been shown to reproduce self-consistent QED screening results for alkali-like systems to within a few percent [5]. These approximations are adequate for the intended purpose of assessing whether QED matters for a given calculation, but PyQED should not be used as a benchmark QED code. The accuracy of the Welton screening model can be assessed by comparing screened and unscreened results. For Th^{4+}, the unscreened point-nucleus Uehling correction is -251.68 eV, while the fully screened finite-nucleus result is -249.29 eV - a combined reduction of 0.9%. In absolute terms, screening and finite nuclear size reduce the total by 2.4 eV, of which most originates from the 1s orbital. A few-percent uncertainty in the Welton model therefore translates to $\sim$0.1 eV absolute error in the total correction. However, because this error is a property of the atomic core, it vanishes in energy

differences with the same completeness as the correction itself and does not affect conclusions about reaction energetics. To confirm this, we verified that the screening correction itself (2.4 eV) is constant across ThF_4, ThF_6^{2-}, and ThF_7^{3-} to within 0.005 eV, ensuring that any error in the Welton model also vanishes in energy differences. For the F^- anion, screening and finite nuclear size effects are negligible ($< 0.01\%$). Since the screening correction is small compared to the total QED effect, and the total QED effect is itself negligible for reaction energetics, the approximate nature of the Welton model does not affect the paper's conclusions. Although PyQED computes only the Uehling (vacuum polarization) contribution, the magnitude of the omitted electron self-energy can be estimated from published data. For atoms in the range Z = 79-111, the self-energy contribution to orbital energies is roughly 4-5 times larger in magnitude than the vacuum polarization, with opposite sign [5, 7]. The ratio decreases weakly with Z as the VP grows faster than SE at high Z [5] Extrapolating to thorium ($Z = 90$), the total self-energy contribution is estimated at roughly +1100 eV with an uncertainty of approximately ±200 eV from the interpolation. This estimate is obtained by interpolating the SE/VP ratios from Schwerdtfeger *et al.*[7] (Au, $Z = 79$: SE/VP ≈ 4.6 for the 1s orbital) and Indelicato *et al.*[5] (element 111, $Z = 111$: SE/VP ≈ 4.4 for the 1s level), giving us an estimated ratio of $\sim$4.5 at $Z = 90$. We note that PyQED reports only the VP component; the net QED correction (VP + SE) is positive for the total atomic energy. The combined QED correction (SE + VP) for the thorium atom is therefore approximately +850 eV. However, because the self-energy is also a core-electron property confined to the nuclear vicinity, it vanishes in energy differences between chemical species the same way as the vacuum polarization. The net self-energy contribution to the ThF_4 atomization energy can be estimated at $\sim$0.5 kcal/mol by the same scaling, which is still well below DFT functional errors of 5-20 kcal/mol. The total QED correction to reaction energetics (including both VP and SE) is therefore expected to be below 1 kcal/mol for actinide fluorides.

D. Future Work

Several extensions would make PyQED more useful. Electron self-energy corrections, the second-largest QED contribution, could be implemented using effective potentials or model-operator approaches to provide complete leading-order QED corrections. Support for additional quantum chemistry packages beyond ORCA (Gaussian, Psi4, Q-Chem) would require only new interface modules, as the QED evaluation is independent of the electronic structure method. Extension to two-component relativistic methods (X2C, DKH) and four-component wavefunctions from DIRAC would improve accuracy for benchmark applications. Analytic gradients of the QED correction with respect to nuclear coordinates would allow assessment of QED effects on equilibrium geometries and vibrational frequencies, though we expect these to be negligibly small. Finally, a systematic study across the actinide and lanthanide series would map the boundary where QED corrections begin to approach chemical accuracy.

VI. CONCLUSIONS

We have presented PyQED, an open-source Python tool that computes Uehling vacuum polarization corrections from converged ORCA calculations. PyQED reads molecular orbital data via the `orca_2json` interface, evaluates the screened finite-nucleus Uehling potential on a Becke integration grid, and returns the QED energy correction as a first-order perturbative estimate. The results quantitatively justify the common practice of neglecting QED in molecular calculations on actinide systems. While QED corrections to total energies are substantial, they transfer unchanged between chemical environments and vanish in energy differences. PyQED makes it straightforward to verify this cancellation for any system of interest, replacing assumption with computation. The code is available under the MIT license at `https://github.com/gbogolyubov/PyQED`.

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